

Boolean Rule Language Compiler Test Cases

Valid Test Cases:

Simple Comparison

```
age := 21;  
adult := age >= 18;  
print adult;
```

Boolean Assignment

```
ready := true;  
print ready;
```

AND NOT Precedence

```
approved := income > 5000 and not blocked;  
print approved;
```

Arithmetic Inside Comparison

```
valid := (score + bonus) >= 60 and attempts < 3;  
print valid;
```

OR AND Precedence

```
flag := true or false and not false;  
print flag;
```

Multiple Assignments

```
base := 10;  
limit := base * 2 + 5;  
allowed := limit >= 20 and not false;  
print allowed;
```

Nested Parentheses

```
status := not ((points + bonus) < threshold) or override;  
print status;
```

Complex Expression

```
eligible := ((income + savings) > debt) and ((age + 1) >= 18 or guardian);  
print eligible;
```

Team 7

Sample

```
ziad := 100000;  
base := ziad / 5;  
allowed := (base * 2 + 5) >= 20 and not false;  
print allowed;
```

Invalid Test Cases:

Missing Semicolon

```
adult := age >= 18  
print adult;
```

Missing Identifier After Print

```
print ;
```

Missing Assignment Expression

```
ready := ;
```

Unclosed Parenthesis

```
valid := (score + bonus >= 60;  
print valid;
```

Dangling AND Operator

```
approved := income > 5000 and ;  
print approved;
```

Expression Starts With Operator

```
flag := or true;  
print flag;
```

Missing RHS Of Comparison

```
adult := age >= ;  
print adult;
```

Double Semicolon

```
ready := true;;  
print ready;
```

Adjacent Operands

```
value := age 18;  
print value;
```

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Invalid Character

```
scoreOk := score @ 60;
```

```
print scoreOk;
```

Malformed Operator

```
check := age <> 18;
```

```
print check;
```

Keyword As Identifier

```
and := true;
```

```
print and;
```